



西安交通大学
XI'AN JIAOTONG UNIVERSITY

**Survival Analysis of Breast Cancer Patients Based on
Regularization Methods**

College: School of Economics and Finance

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Class: S5192

Name: Peng Xinyi

Student ID: 3125319148

Supervisor: Prof. Jiang He

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ABSTRACT

Breast cancer is a highly heterogeneous disease, and patient prognosis often varies significantly even within established molecular subtypes, presenting a major challenge for precise risk stratification. This study leverages the METABRIC cohort, comprising 1,904 breast cancer patients, to construct a prognostic model for survival prediction from high-dimensional clinical and genomic data. To address the statistical challenges of dimensionality and multicollinearity, we employed regularized Cox proportional hazards models (Lasso, Ridge, ElasticNet) for their ability to stabilize coefficient estimates and perform feature selection. Following data preprocessing, which refined the dataset to 901 samples and 640 features, model performance was evaluated via the C-index. The analysis revealed that while all regularized models demonstrated comparable discriminative ability (C-index 0.70-0.71), ElasticNet offered a superior balance by selecting a robust subset of 41 features. To further enhance predictive performance, ensemble methods like LightGBM were applied to the pre-selected feature set, achieving a top C-index of 0.721. Critically, all models consistently identified lymph node status, tumor size, tumor stage, and specific genes including *CXCL8* and *GATA3_mut* as the most potent prognostic factors. This study validates the utility of regularization for deriving interpretable models from high-dimensional oncological data and identifies a concise set of clinical and molecular features with significant potential to improve risk stratification in breast cancer.

KEY WORDS: Breast Cancer; Regularization; Cox; Machine Learning

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1 Introduction

1.1 Background Analysis

Breast cancer remains one of the most prevalent and heterogeneous malignancies worldwide, posing significant challenges to clinical prognosis and personalized treatment strategies^[1]. Despite advances in molecular subtyping systems such as PAM50, which classify breast tumors into intrinsic subtypes like Luminal A, Luminal B, and HER2-enriched, substantial variability in patient outcomes persists even within the same subtype. This prognostic ambiguity underscores the limitations of current classification frameworks and highlights the urgent need to identify more granular molecular markers that can refine risk stratification. The emergence of high-throughput genomic technologies has enabled the generation of large-scale molecular datasets, integrating gene expression profiles, mutation status, and clinical variables.

However, the high-dimensional nature of such data, often characterized by a large number of features relative to sample size, introduces statistical challenges, including overfitting, multicollinearity, and difficulty in interpretability. These issues complicate the development of robust prognostic models that are both predictive and clinically actionable.

1.2 Contributions and Outline of the Paper

To address these challenges, this study leverages the METABRIC cohort, a comprehensive dataset comprising clinical attributes, mRNA expression levels, and mutation profiles from 1,904 breast cancer patients. We employ regularization methods—Lasso, Ridge, and ElasticNet—within a Cox proportional hazards framework to build a multivariable prognostic model that integrates clinical and genomic features. The objective is to enhance predictive accuracy while maintaining interpretability, thereby bridging the gap between high-dimensional data analysis and clinical utility. Furthermore, we extend our analysis to ensemble learning methods, including random survival forests and gradient-boosting models, to capture potential non-linear relationships and interactions among features.

The remainder of this paper is organized as follows: Section 2 describes the data source and preprocessing steps; Section 3 outlines the methodological framework; Section 4 presents exploratory data analysis; Section 5 details model construction and evaluation; Section 6 discusses further analyses with ensemble methods; and Section 7 concludes with clinical implications and future directions.

2 Data Description and Preprocessing

2.1 Data Source

The data utilized in this study was sourced from the Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) project, a landmark cohort study jointly propelled by The Cancer Genome Atlas (TCGA) and the Cancer Research UK Cambridge Institute^[2]. This comprehensive dataset, publicly accessible via the cBioPortal^[3] for Cancer Genomics and subsequently hosted on Kaggle[®], integrates detailed clinical information, gene expression profiles, and mutation data, making it an invaluable resource for breast cancer research^[4]. The original dataset comprised 1,904 patient samples and 692 feature variables, characterizing it as a high-dimensional dataset.

2.2 Variable Description

The variables can be broadly categorized into three distinct groups: clinical features, mRNA expression data and somatic mutation data.

2.2.1 Clinical Features

The first group encompasses clinical features, which include 30 variables detailing patient demographics, tumor pathology, and treatment history. Key identifiers and demographic variables include `patient_id` and `age_at_diagnosis`. Pathological characteristics are captured by variables such as `tumor_size`, `tumor_stage`, `lymph_nodes_examined_positive`, `neoplasm_histologic_grade`, and `pam50_intrinsic_subtypes`. Treatment information includes the type of breast surgery (`type_of_breast_surgery`), chemotherapy, hormone therapy, and radiotherapy. Critically, the survival outcomes are defined by `overall_survival_months` (the time-to-event variable) and `overall_survival_status` and `death_from_cancer` (the event indicator variables).

2.2.2 mRNA Expression Data

The second group consists of mRNA expression data, comprising 489 features. These values represent the standardized expression levels of individual genes. The data had already undergone Z-score normalization prior to public release, ensuring that for each gene, the expression values across all samples have a mean of zero and a standard deviation of one, thereby facilitating comparative analysis between genes.

[®] Dataset in Kaggle: <https://www.kaggle.com/datasets/raghadalharbi/breast-cancer-gene-expression-profiles-metabric/data>

2.2.3 Somatic Mutation Data

The third group contains somatic mutation data, recording the presence of specific genetic alterations in 173 genes. These data, often represented in a binary or categorical manner (e.g., wild-type vs. mutant, or specific mutation types like H1047R in PIK3CA), provide insight into the genomic drivers of tumorigenesis. The integration of these three data types offers a multi-faceted view of each tumor, enabling a holistic approach to prognostic model development.

2.3 Data Preprocessing

A rigorous multi-step preprocessing pipeline was implemented to ensure data quality and model robustness, given the high-dimensional nature of the dataset. The initial dataset of 1,904 samples was subsequently refined to a final cohort of 901 samples through this process.

2.3.1 Missing Value Handling

The first step in data preprocessing is missing value analysis. Preliminary results of missing value statistics show that, only clinical characteristic variables were missing. The top five variables with the most missing values were `tumor_stage`, `3-gene-classifier-subtype`, `primary_tumor_laterality`, `neoplasm_histologic_grade` and `cellularity`. The statistics and waterfall plots for each missing value are shown in the figure 2-1 and figure 2-2.

Different types of missing values require different preprocessing methods. Common missing value patterns include missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR). Therefore, a heatmap of correlations between missing variables was first used to examine the missing value patterns, which is shown in figure 2-3. The results in the figure 2-3 show a complete co-missing pattern among the three variables related to cancer subtypes (`oncotree_code`, `cancer_type_detailed` and `tumor_other_histologic_subtype`), indicating complete collinearity. This suggests systematic missing values, possibly due to unclassified cancer subtypes; therefore, redundant variables were directly removed. For the remaining variables, based on the correlation coefficient, the missing data was determined to be randomly missing; therefore, missing values were directly removed, resulting in a final sample size of 1092.

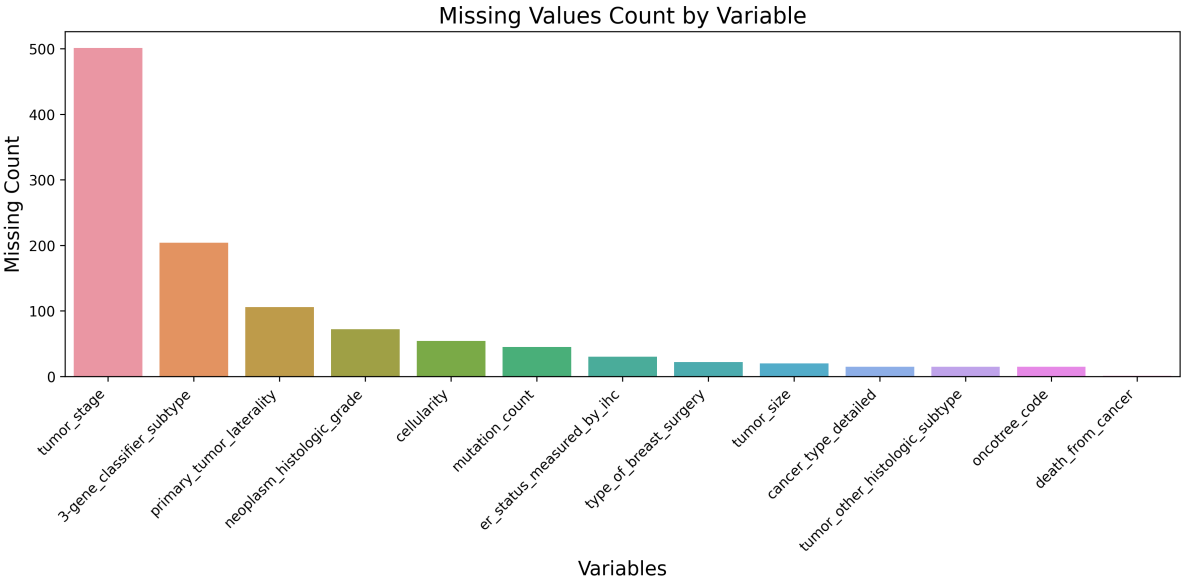


Figure 2-1 Missing Value Count

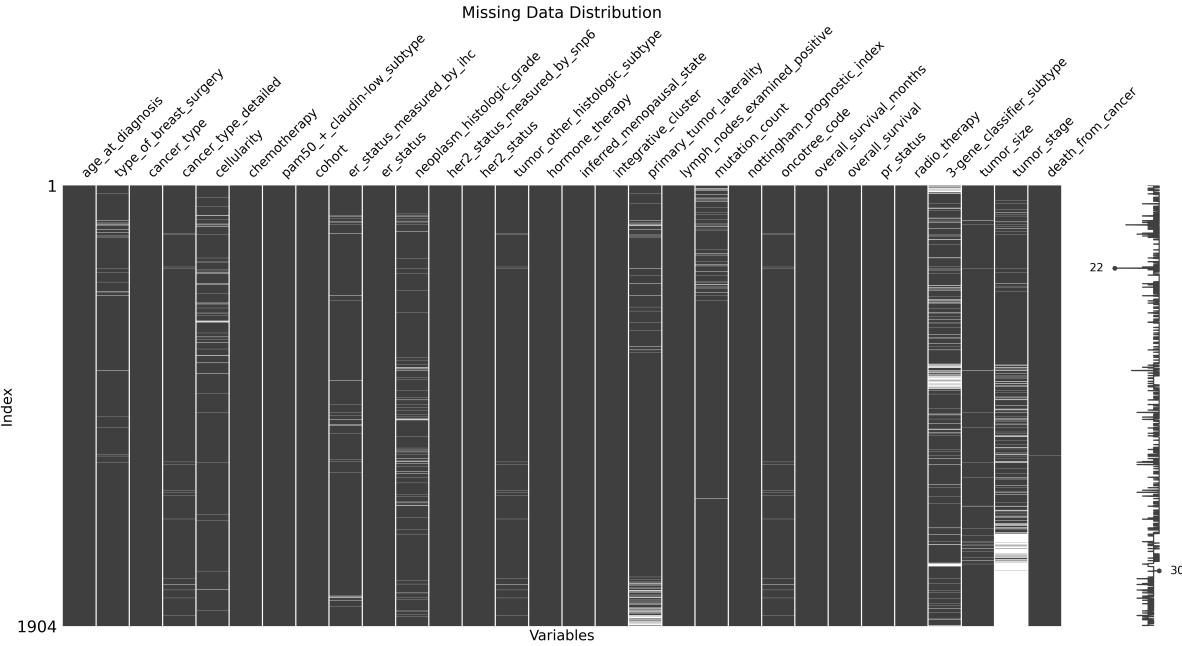


Figure 2-2 Missing Value Distribution

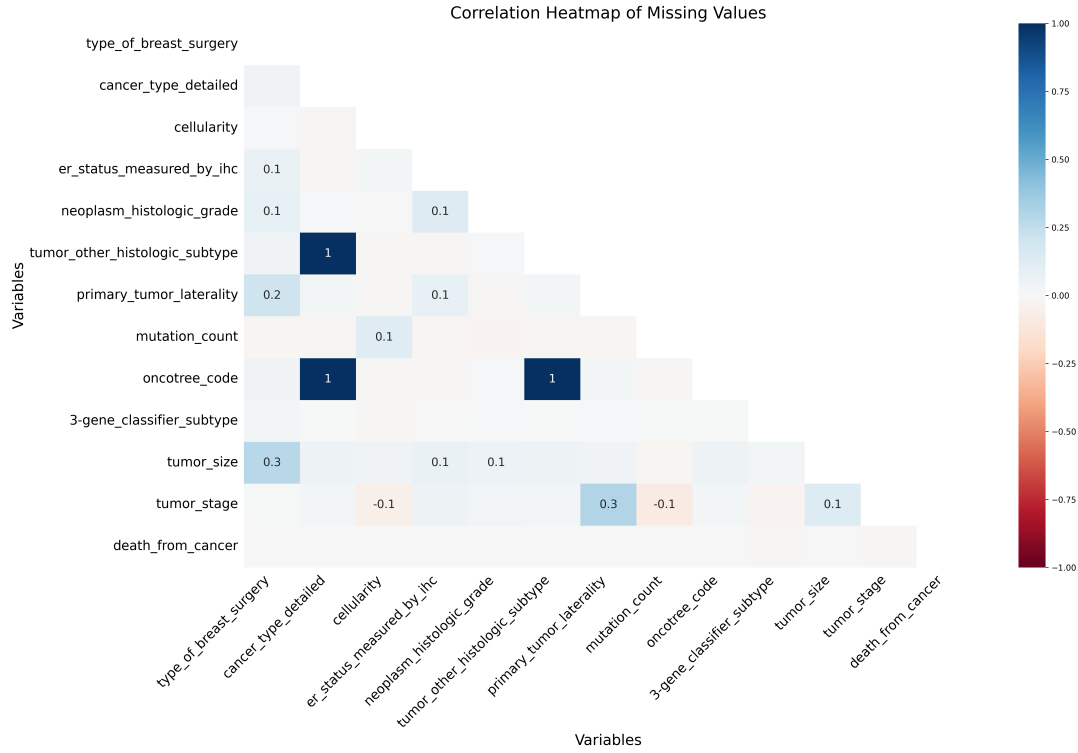


Figure 2-3 Correlation Heatmap of Missing Variables

2.3.2 Outlier Analysis and Treatment

For continuous clinical variables, outlier detection was performed using boxplot visualization. From figure 2-4, variables such as tumor_size, mutation_count, and lymph_nodes_examined_positive were found to contain significant outliers.

To prevent these extreme values from unduly influencing the model, all samples containing outliers were removed, resulting in a cleaned dataset of 901 samples.

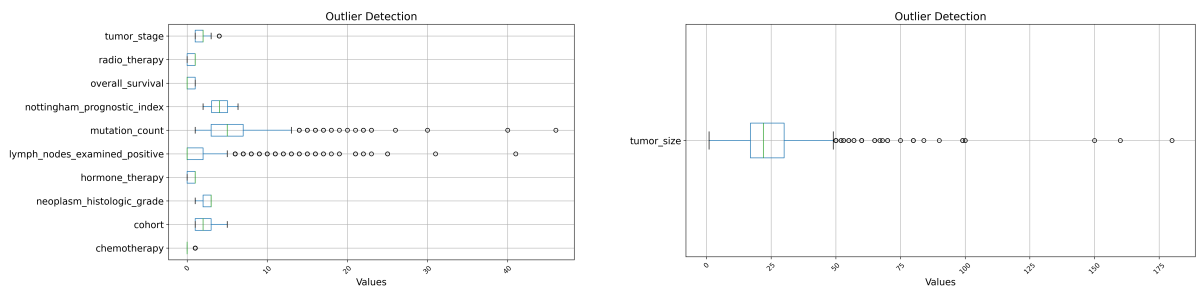


Figure 2-4 Outlier Detection (Before)

Figure 2-5 shows the box plot results after outlier removal, indicating that the outlier problem has been effectively mitigated.

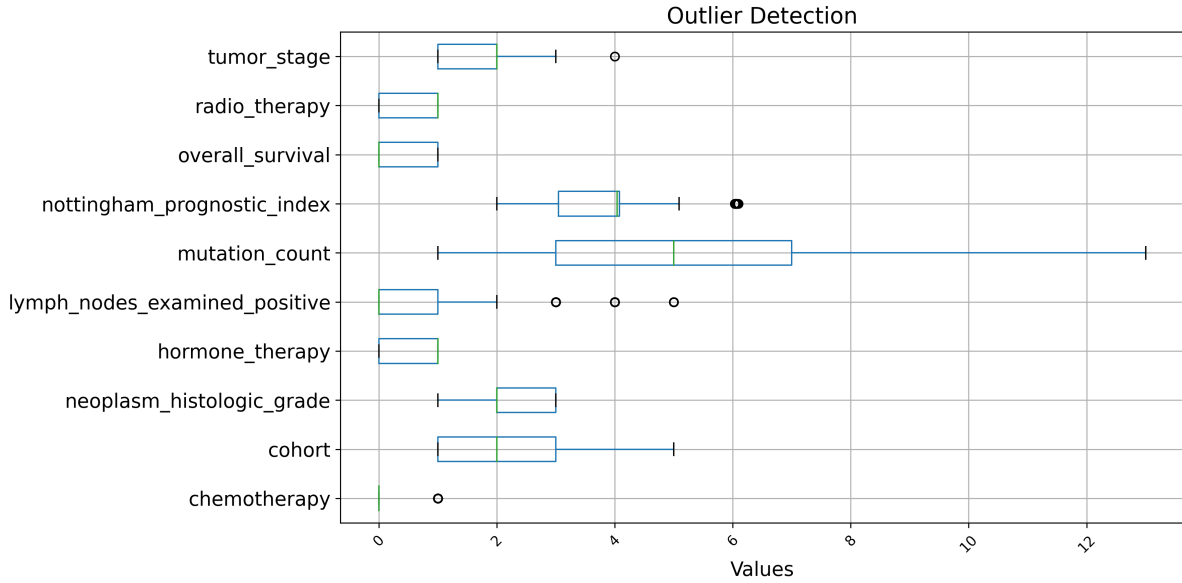


Figure 2-5 Outlier Detection (After)

2.3.3 Feature Engineering

Feature engineering consists of two parts: First of all, categorical clinical variables were transformed into a numerical format suitable for modeling using one-hot encoding. This process expanded the feature space, resulting in an intermediate data shape of (901, 722).

Secondly, the current gene mutation characteristics are still a mixed data type (no mutations are marked as 0, and mutated mutations are listed with their specific mutation types). Since the mutation rate at each gene locus is low, however, the number of mutation types can reach dozens, one-hot encoding of each mutation type at each locus would further improve data sparsity, but with little effect and increased computational cost. Therefore, we directly mark all non-zero values as 1, indicating that a mutation has occurred at a certain gene locus in the patient.

2.3.4 Collinearity Check

Given the high dimensionality, assessing multicollinearity was critical. Even though regularization methods can also reduce the dimensionality of data, the preliminary collinear variable removal can ensure the smooth operation of subsequent matrix calculations and reduce computational complexity.

The Variance Inflation Factor (VIF) was employed as a diagnostic measure. The core idea of VIF is to use one independent variable to regress all other independent variables and see how well the regression model fits. If the fit is very good (i.e., R^2 is high), it means that this independent variable can be "linearly explained" by the other independent variables, which implies strong collinearity among them. For a specific independent variable X_j , its VIF is calculated using the following formula:

$$VIF_j = \frac{1}{1 - R_j^2} \quad (2-1)$$

In the equation 2-1, VIF_j is the variance inflation factor of variable X_j , R^2 is the coefficient of determination obtained by regressing X_j as the dependent variable against all other independent variables. Generally, a VIF greater than 10 can be considered as indicating severe multicollinearity.

The initial VIF calculation revealed severe multicollinearity, with numerous features, particularly those related to detailed cancer subtypes and receptor status, exhibiting VIF values exceeding the threshold of 10, with some being infinite. Therefore, an iterative process was undertaken, sequentially removing the variable with the highest VIF until all remaining variables had a VIF below 10. This procedure eliminated 72 highly collinear features, resulting in a refined dataset with 650 features.

2.3.5 Model Data Preparation

Finally, we move on to the data preparation stage. The final processing steps include four parts.

Firstly, continuous variables were standardized to have a mean of zero and a standard deviation of one, using the following formula:

$$x_{\text{standardized}} = \frac{x - \mu}{\sigma} \quad (2-2)$$

Secondly, define the features as well as the target variables. Firstly, the features are the three types of feature variables mentioned above: clinical variables, mRNA expression data, and somatic mutation features. As for target variables in survival analysis, it is somehow different with common regression or classification task. The target variable for the survival analysis was a tuple-like variable, containing two variables: time and event. In this study, time refers to the total survival months during the observation period, using the variable of `overall_survival_month`. For the event variable, considering that this study primarily focuses on survival factors in breast cancer, thus, using the variable of `overall_survival`, and death due to breast cancer is defined as an event occurring (assigned a value of 1), while surviving patients, or those who died from other diseases or were lost to follow-up, were considered as not having an event occurring (assigned a value of 0), treated as censor samples.

Thirdly, features with near-zero variance (variance < 0.1) were removed, as they contribute little predictive information and can increase model complexity. This step eliminated 10 additional variables.

Finally, the dataset (now 901 samples, 640 features) was split into training and testing sets using an 8:2 ratio. To preserve the distribution of event rates in both sets, the split was performed using stratified sampling based on the event indicator, yielding a training set of 720 samples (266 events) and a test set of 181 samples (67 events).

At this point, we have already completed the data preprocessing of this study, and we can now proceed to the data exploration and model analysis phases.

3 Exploratory Data Analysis

3.1 Distribution of Continuous Variables

An exploratory data analysis was conducted to understand the underlying distributions of key variables, assess the relationships between treatments and survival, and identify significant prognostic factors through univariable survival analysis.

The univariate analysis of continuous clinical variables revealed several key patterns. The age at diagnosis followed an approximately normal distribution, with a peak incidence observed in the 55-65 year age range. The distribution of tumor size indicated that the majority of tumors were between 2 cm and 5 cm in diameter, which aligns with typical clinical presentations. Regarding lymph node status, a critical prognostic factor, approximately 35% of the patients in the cohort had positive lymph node involvement, indicating regional metastasis. The mutation count per patient exhibited a right-skewed distribution, suggesting that while most patients had a lower burden of mutations, a subset harbored a significantly higher number of genetic alterations. Finally, the distribution of overall survival time was also right-skewed, a common characteristic of survival data where a considerable number of patients are censored, and a long tail represents long-term survivors.

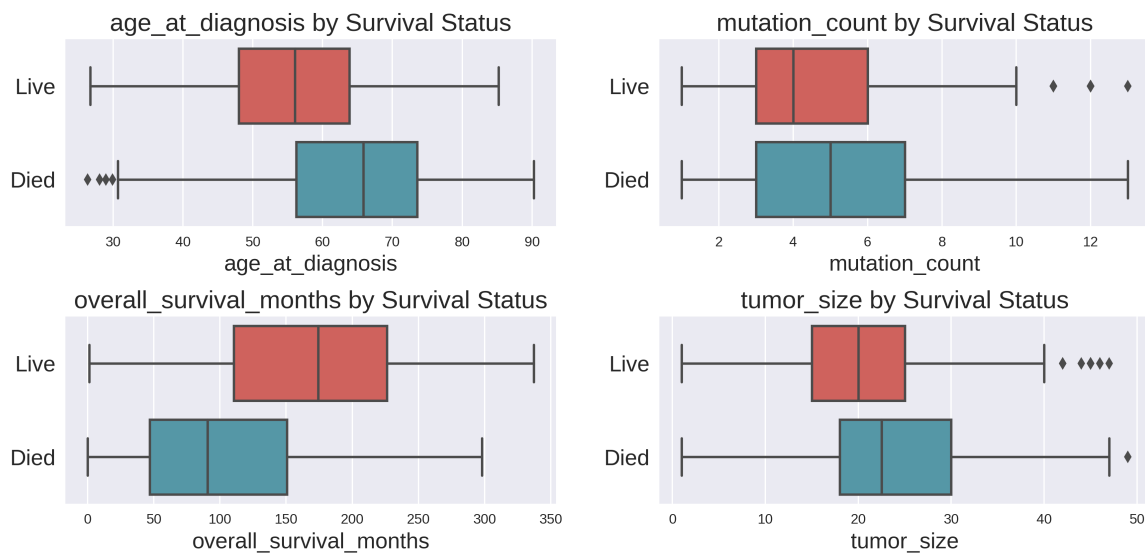


Figure 3-1 Distribution of Continuous Variables

3.2 Analysis Based on Therapy Type

In the clinical treatment of breast cancer, chemotherapy, hormone therapy, and radiotherapy are the three main treatment methods. Patients choose the appropriate treatment plan based on their own conditions. First, a descriptive statistical analysis was performed on the number of patients who chose each treatment plan, results are shown in figure. Based on the results we can see that chemotherapy was the most frequently administered intervention in this cohort, followed by hormone therapy and radiotherapy.

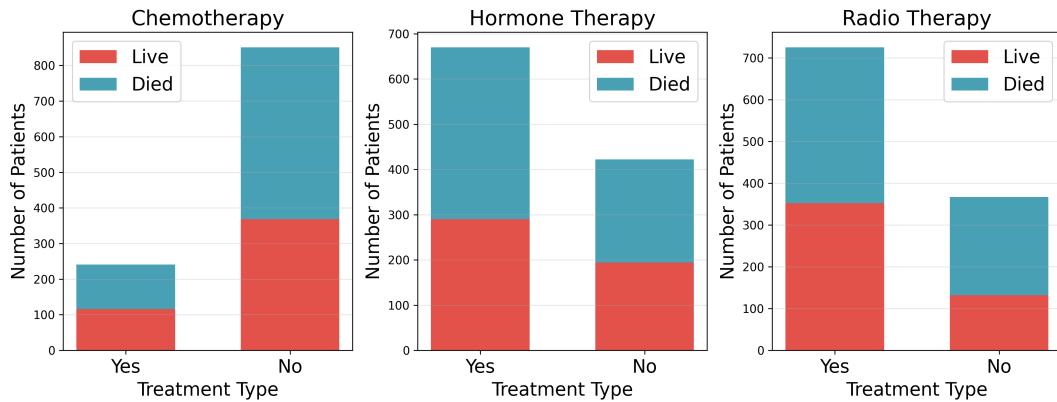


Figure 3-2 Summary of Three Therapy Type

The relationship between therapy and patient survival was further investigated. We used the Kaplan-Meier curve, a commonly used analytical method in the field of survival analysis, to reveal survival differences between different patient groups. The results showed significant separation of the KM curves in all three figures. This indicates that all three treatment regimens demonstrated a significant survival benefit compared to the untreated group. Based on the degree of curve separation, the overall survival benefit was chemotherapy > hormone therapy > radiotherapy. Notably, although patients receiving chemotherapy had a slightly lower initial survival probability (possibly due to the acute toxicity and side effects of chemotherapy), the long-term survival advantage of the chemotherapy group became significant over time. The radiotherapy and hormone therapy groups maintained relatively stable survival advantages throughout the observation period.

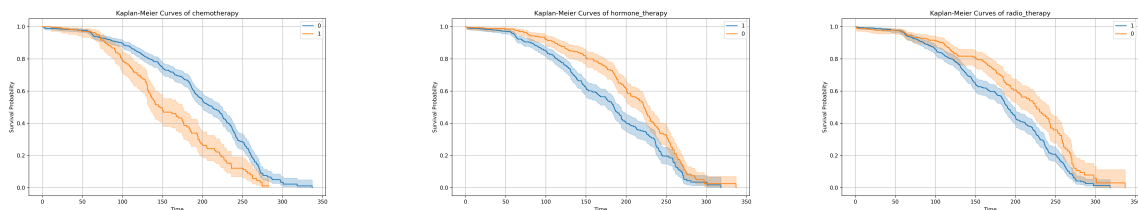


Figure 3-3 KM Curve of Different Therapy

3.3 Single Factor KM Curve

To evaluate the prognostic power of individual categorical clinical features, Kaplan-Meier survival analysis was performed for variables including tumor_stage, 3-gene_classifier_subtype, cellularity, and type_of_breast_surgery. The resulting survival curves were examined for their degree of separation.

According to the results in figure ,features such as tumor stage and molecular subtype demonstrated pronounced stratification, with distinct and well-separated survival curves, confirming their strong univariable association with patient outcomes. This initial analysis helped to validate known clinical prognostic factors and provided a foundation for the subsequent multivariable modeling.

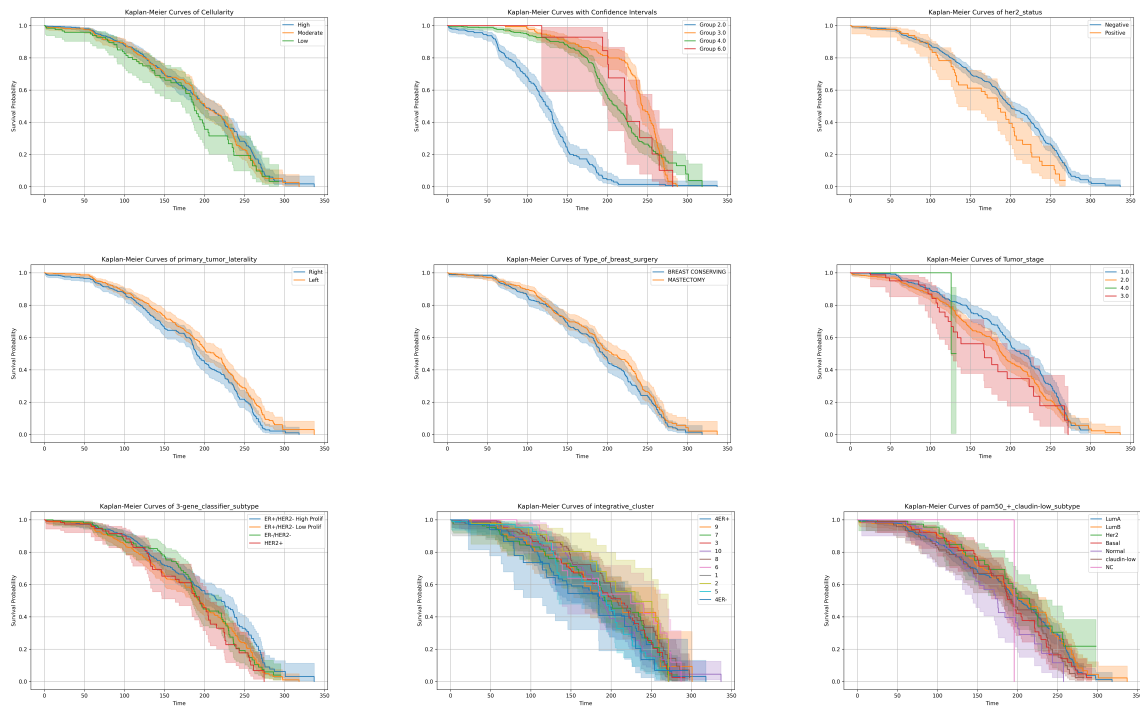


Figure 3-4 KM Curve of Different Single Variables

4 Methodology

4.1 Cox Proportional Hazards Model

4.1.1 Model Structure

The Cox proportional hazards (PH) model is a semi-parametric approach widely used in survival analysis to assess the effect of covariates on the hazard of an event occurring. The model specifies the hazard function for an individual at time t with a covariate vector $\mathbf{X}$ as:

$$h(t|\mathbf{X}) = h_0(t)\exp(\boldsymbol{\beta}^T \mathbf{X}) = h_0(t)\exp\left(\sum_{j=1}^p \beta_j X_j\right) \quad (4-1)$$

where $h_0(t)$ is the non-parametric baseline hazard function, representing the hazard when all covariates are zero, and $\boldsymbol{\beta} = (\beta_1, \beta_2, \dots, \beta_p)^T$ is the vector of regression coefficients to be estimated. The model is termed "proportional" because the hazard for any two individuals is assumed to be constant over time.

Parameter estimation in the Cox model is performed by maximizing the partial likelihood function, which does not require specification of the baseline hazard $h_0(t)$. For n independent observations, the data is denoted as $\{t_i, \delta_i, \mathbf{X}_i\}$, where t_i is the observed time, δ_i is the event indicator (1 if the event occurred, 0 if censored), and $\mathbf{X}_i$ is the covariate vector for the i -th individual. The risk set $R(t_i)$ is the set of individuals still at risk (alive and uncensored) at time t_i . The partial likelihood function is:

$$L(\boldsymbol{\beta}) = \prod_{i:\delta_i=1} \frac{\exp(\boldsymbol{\beta}^T \mathbf{X}_i)}{\sum_{j \in R(t_i)} \exp(\boldsymbol{\beta}^T \mathbf{X}_j)} \quad (4-2)$$

Maximizing this likelihood is equivalent to minimizing the negative log-partial likelihood:

$$\ell(\boldsymbol{\beta}) = -\log L(\boldsymbol{\beta}) = -\sum_{i:\delta_i=1} \left[\boldsymbol{\beta}^T \mathbf{X}_i - \log \left(\sum_{j \in R(t_i)} \exp(\boldsymbol{\beta}^T \mathbf{X}_j) \right) \right] \quad (4-3)$$

4.1.2 Evaluation

The Cox proportional hazards model's predictive performance was evaluated using the Concordance Index (C-index), a generalization of the area under the ROC curve (AUC) for survival data that accounts for censoring. The C-index quantifies the model's ability to produce a reliable ranking of individuals by their risk of experiencing the event.

The core principle involves evaluating all possible comparable pairs of individuals in the dataset. For a pair of individuals (i, j) , it is considered comparable if the one with the shorter observed survival time experienced the event (i.e., was not censored). Formally, a pair is comparable if $t_i < t_j$ and $\delta_i = 1$, where t is the observed time and δ is the event indicator.

For each such comparable pair, the model's predicted risk scores (typically the linear predictor $\hat{\eta} = \mathbf{X}\hat{\beta}$ from the Cox model) are compared to the actual outcomes:

- The pair is concordant if the individual with the higher risk score (e.g., $\hat{\eta}_i > \hat{\eta}_j$) is the one who experienced the event earlier (i.e., $t_i < t_j$).
- The pair is discordant if the individual with the higher risk score has a longer survival time.
- The pair is tied if the risk scores are equal.

The C-index is then calculated as the proportion of concordant pairs among all comparable pairs, see equation 4-4.

$$\text{C-index} = \frac{\text{Number of Concordant Pairs} + 0.5 \times \text{Number of Tied Pairs}}{\text{Number of Comparable Pairs}} \quad (4-4)$$

The value of the C-index ranges from 0 to 1. A value of 0.5 indicates that the model's predictions are no better than random chance. A value of 1.0 signifies perfect discrimination, meaning every comparable pair is correctly ranked by the model. In clinical and genomic prognostic studies, a C-index above 0.7 is typically considered to indicate a model with useful predictive ability, while a value above 0.8 represents strong discrimination. It is important to note that the C-index is a measure of rank correlation between predicted risk and observed survival times and does not measure the absolute accuracy of the predicted survival probabilities.

4.2 Regularization

In high-dimensional settings where the number of features p is large relative to the sample size n , standard maximum partial likelihood estimation becomes unstable or infeasible.

Regularization methods address this by adding a penalty term to the loss function to constrain the magnitude of the coefficients, prevent overfitting, and, in some cases, perform feature selection. The general form of the objective function of the Cox proportional hazards model with a penalty term can be expressed as equation 4-5.

$$\min_{\beta} \{ \ell(\beta) + \lambda P(\beta) \} \quad (4-5)$$

where $\ell(\beta)$ is the negative log-partial likelihood, $\lambda \geq 0$ is a tuning parameter controlling the strength of the penalty, and $P(\beta)$ is the penalty function.

In this study, we employ three types of regularization: Lasso, Ridge and ElasticNet.

4.2.1 Lasso

The Least Absolute Shrinkage and Selection Operator (Lasso) uses an L1-norm penalty: $P(\beta) = \|\beta\|_1 = \sum_{j=1}^p |\beta_j|$. This penalty has the property of shrinking some coefficients exactly to zero, thereby performing automatic feature selection. By applying a lasso regularization to the Cox model, a Lasso-Cox model is constructed. The objective function of this model can be adjusted as follows:

$$\min_{\beta} \left\{ \ell(\beta) + \lambda \sum_{j=1}^p |\beta_j| \right\} \quad (4-6)$$

Lasso's main advantage lies in its powerful feature selection capability. By compressing the coefficients of unimportant variables to zero through an L1 penalty term, it can automatically generate a sparse model, which greatly enhances the model's interpretability and helps identify the most critical features in high-dimensional data.

However, its core drawback lies in its performance when dealing with highly correlated features: when a set of features is highly correlated, Lasso tends to randomly select one while setting the coefficients of other correlated features to zero, which can lead to model instability. Furthermore, when the number of features is much larger than the number of samples ($p \gg n$), it can only select a maximum of n variables.

4.2.2 Ridge

The Ridge penalty uses an L2-norm: $P(\beta) = \|\beta\|_2^2 = \sum_{j=1}^p \beta_j^2$. This penalty shrinks coefficients towards zero but does not set them exactly to zero. Based on the regularization of the L2-norm, we can build a Ridge-Cox model, the objective function of this model can be expressed as equation 4-7:

$$\min_{\beta} \left\{ \ell(\beta) + \lambda \sum_{j=1}^p \beta_j^2 \right\} \quad (4-7)$$

Ridge regression's main advantage lies in its superior ability to handle multicollinearity. By shrinking the coefficients through an L2 penalty term without setting them to zero, it stabilizes the coefficient estimates, making the model less sensitive to small changes in the data and thus effectively improving generalization ability.

Its disadvantage is the lack of an inherent feature selection mechanism. It retains all features in the model, resulting in poor interpretability of the final model, especially in high di-

mensional scenarios where it is difficult to identify the most critical risk factors from hundreds of non-zero coefficients.

4.2.3 ElasticNet

The ElasticNet penalty is a convex combination of the L1 and L2 penalties: $P(\beta) = \alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2$. It aims to balance the feature selection capability of Lasso with the stability and group effect handling of Ridge regression. Through this we can build a ElasticNet-Cox model, the objective function is:

$$\min_{\beta} \left\{ \ell(\beta) + \lambda \left(\alpha \sum_{j=1}^p |\beta_j| + \frac{1-\alpha}{2} \sum_{j=1}^p \beta_j^2 \right) \right\} \quad (4-8)$$

ElasticNet have two hyperparameters. In addition to adjusting the penalty coefficient λ , we also need to adjust the coefficient α to balance the strengths of the two regularization. When $\alpha = 1$, the ElasticNet is completely equivalent to Lasso, possessing the strongest feature selection capability. When $\alpha = 0$, the ElasticNet is completely equivalent to Ridge, lacking any feature selection capability.

Overall, ElasticNet combine the advantages of both regularization methods, handling group effects while selecting features, and are more stable than Lasso. However, the parameter tuning process becomes more complex and computationally expensive.

4.3 Ensemble Learning Models

To capture potential non-linear relationships and complex interactions among features that may not be fully captured by the linear Cox PH model, this study also employed ensemble learning methods.

4.3.1 Random Survival Forest (RSF)

RSF is an extension of Random Forests for survival data. It builds an ensemble of survival trees. At each node split, a candidate set of features is considered, and the split is chosen to maximize the difference in survival between the resulting daughter nodes, often measured by the log-rank statistic. The ensemble's prediction is an aggregated survival function. RSF naturally handles non-linearity and interactions and provides measures of variable importance.

4.3.2 XGBoost

Boosting methods combine multiple weak learners sequentially, where each new tree corrects the errors of the combined ensemble from the previous steps.

XGBoost (Extreme Gradient Boosting) is a highly efficient and scalable implementation

of gradient boosting that uses a second-order approximation of the loss function and regularization to control model complexity. Its objective for survival analysis (Cox loss) includes both the partial likelihood loss and a regularization term $\Omega(f_k)$ for tree k :

$$L_{\text{XGBOOST}} = \sum_{i=1}^n l(\mathbf{y}_i, \hat{\mathbf{y}}_i) + \sum_{k=1}^K \Omega(f_k) \quad (4-9)$$

The regularization term $\Omega(f_k)$ for a tree f_k in XGBoost is given by:

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (4-10)$$

Where T is the number of leaves in the tree; w_j is the score (weight) on leaf j ; γ and λ are regularization parameters that penalize the number of leaves and the magnitude of the leaf weights, respectively.

4.3.3 LightGBM

LightGBM (Light Gradient Boosting Machine) is also a gradient boosting framework designed for efficiency and scalability. It uses two novel techniques: Gradient-based One-Side Sampling (GOSS) to focus on instances with large gradients and Exclusive Feature Bundling (EFB) to handle high-dimensional sparse features effectively. It often achieves comparable or better accuracy than XGBoost with faster training times. LightGBM also optimizes an objective function composed of the Cox loss and a regularization term, which can be approximated as follows:

$$L_{\text{LightGBM}} \approx \sum_{i=1}^n l(\mathbf{y}_i, \hat{\mathbf{y}}_i) + \text{Regularization}(f_k) \quad (4-11)$$

As for regularization, LightGBM similarly controls complexity through parameters like `max_depth`, `num_leaves`, and `min_data_in_leaf`, effectively penalizing complex trees to prevent overfitting.

5 Regularized Cox PH Model

The core of modeling approach involves the implementation of three regularized Cox Proportional Hazards models: Lasso, Ridge, and ElasticNet, to address the high dimensionality of the feature space. The optimal regularization hyperparameter λ for each model was determined via 5-fold cross-validation on the training set, with the goal of maximizing the cross-validated Concordance Index (C-index). The C-index was employed as the primary performance metric to evaluate the models' ability to correctly rank patients by their risk of death.

5.1 Lasso-Cox

The Lasso-Cox model applied an L1-penalty to the Cox partial log-likelihood. The result in figure 4-6 shows that, the cross-validation curve for the partial likelihood deviance against the $\log(\lambda)$ values exhibited a characteristic U-shape, confirming the presence of an optimal penalty strength that balances bias and variance. The optimal λ was selected as 0.0560.

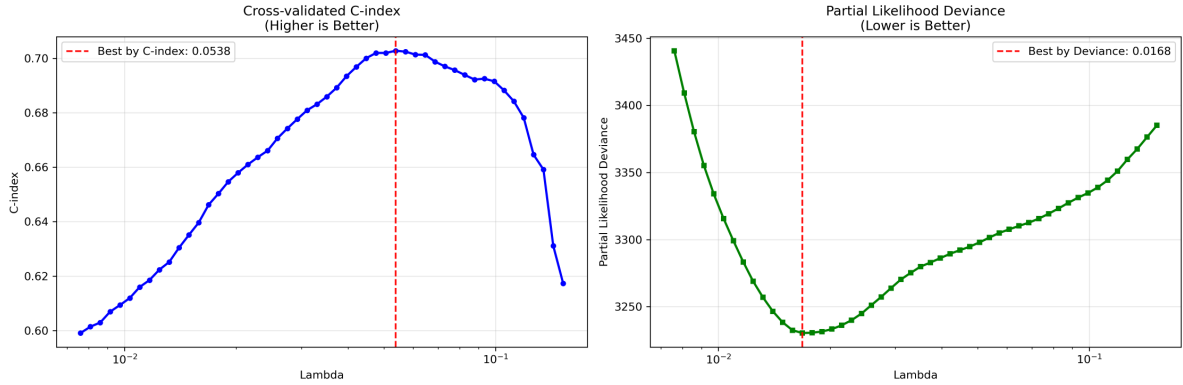


Figure 5-1 Training Results of Lasso-Cox

Lasso-Cox demonstrated a strong feature selection capability. As λ increased, the coefficients of a majority of features were shrunk to zero, as shown in figure 5-2. Ultimately, the Lasso-Cox model identified a highly sparse set of only 18 predictive features (2.8% of the total), while achieving a C-index of 0.72 on the training set and 0.703 on the held-out test set.

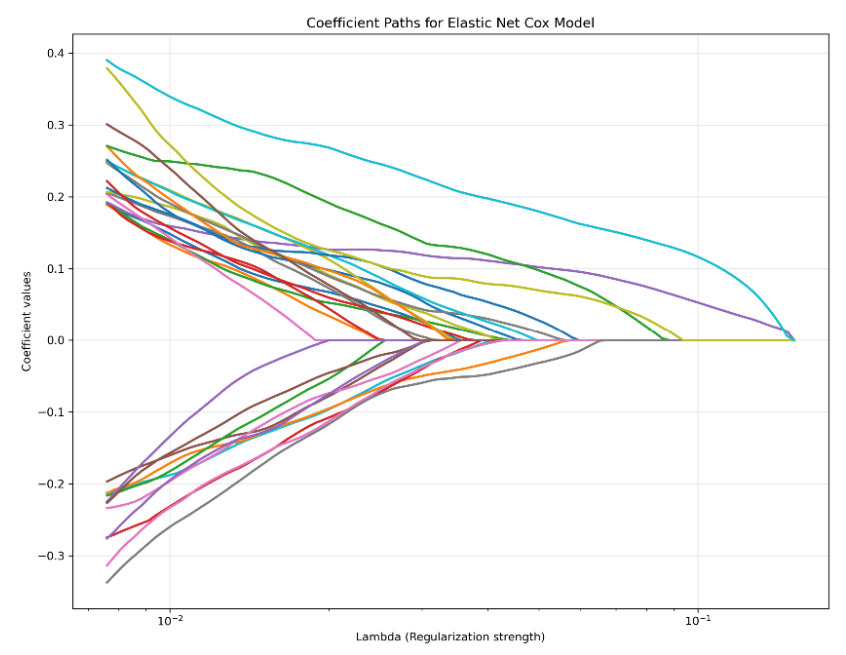


Figure 5-2 Variable Coefficient Changes in Lasso-Cox

Lasso-Cox model demonstrates a strong feature selection capability. Variable importance result shows that (Figure 5-3), clinical features such as lymph_node_status, tumor_size, integrative_cluster_6, and tumor_stage significantly contribute to predicting patient survival risk; genes such as cxc18, ncoa3, gata3_mut, slc9a1, pms2, and smarcd1 also play important predictive roles.

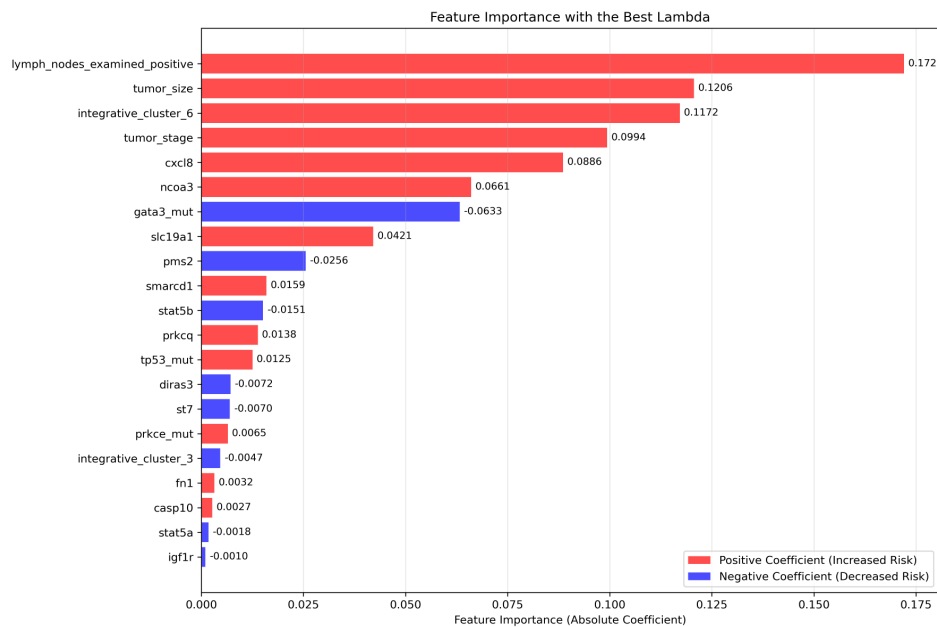


Figure 5-3 Caption

5.2 Ridge-Cox

In contrast, the Ridge-Cox model employs L2 regularization, which shrinks coefficients towards zero, but not completely, thus preserving all features.

As shown in the histogram of variable coefficients in Figure 5-4, the vast majority of coefficients fall within the range slightly greater than 0, reflecting the coefficient compression characteristic of L2 regularization.

Based on five-fold cross-validation, the optimal parameters ultimately selected were $\lambda = 658.39$. Notably, the C-index of Ridge-Cox on the test set is 0.706, slightly higher than the Lasso-Cox.

Further analysis of variable importance, listing the top 20 most important variables, reveals a significant overlap with the Lasso-Cox feature selection results, further confirming the importance of features like lymph_node_status, tumor_size, integrative_cluster_6, and tumor_stage. However, Ridge-Cox also assigns significant importance to other variables, such as PAM50 subtype, tumor histological grade, and chemotherapy, which have been excluded in the more aggressive Lasso regularization, suggesting that these variables may provide additional predictive information. Maybe that's why Ridge-Cox achieved better prediction results than Lasso-Cox.

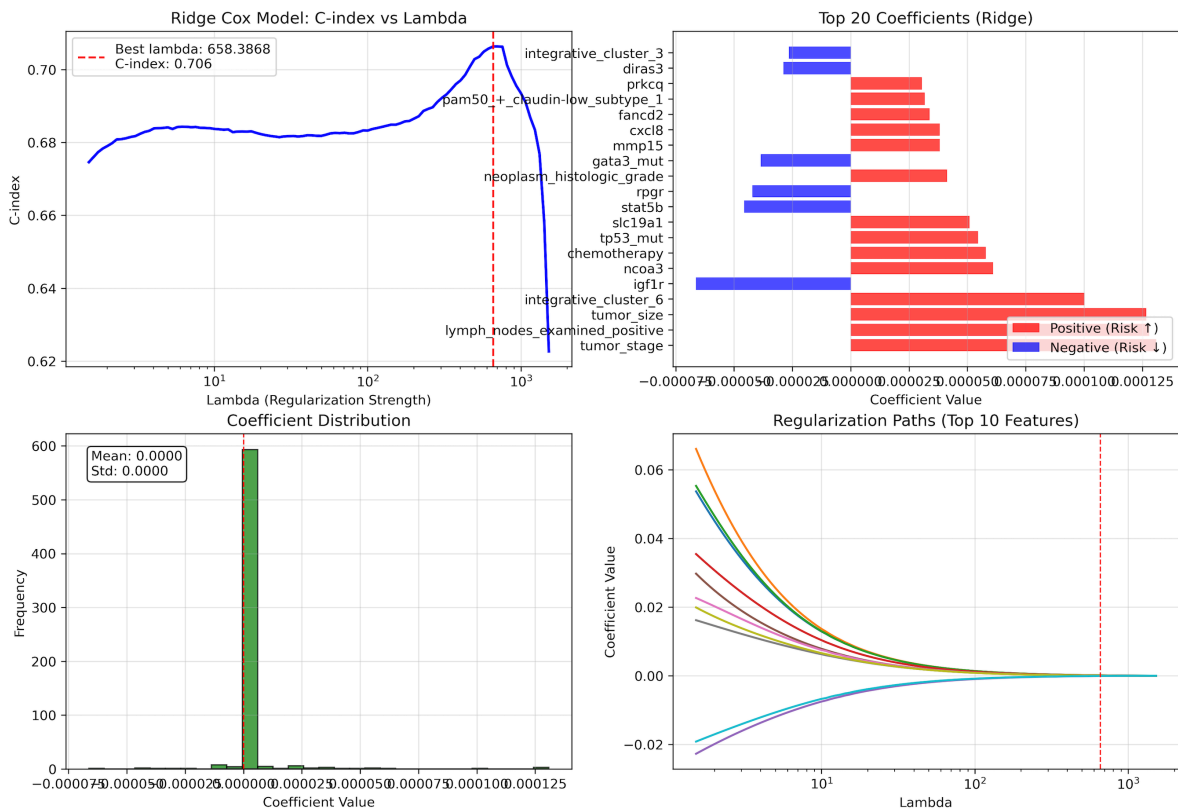


Figure 5-4 Model Result of Ridge-Cox

5.3 ElasticNet

To leverage the advantages of Lasso and Ridge regression, we implemented an ElasticNet-Cox model combining L1 and L2 penalties.

Since ElasticNet-Cox has two hyperparameters, a grid search was first performed on the L1 ratio (α) from 0.1 to 0.9. The relationship between the L1 ratio and the test C-index was analyzed, as shown in figure 5-5, indicating that the model with predominantly L2 penalties (lower L1 ratio) performed best. Based on the optimal L1 ratio of 0.1, five-fold cross-validation was used again to find the optimal solution for λ , got a result of $\lambda = 0.4698$. Based on this, the test C-index reached 0.709, the highest among the three Cox regression-based models.

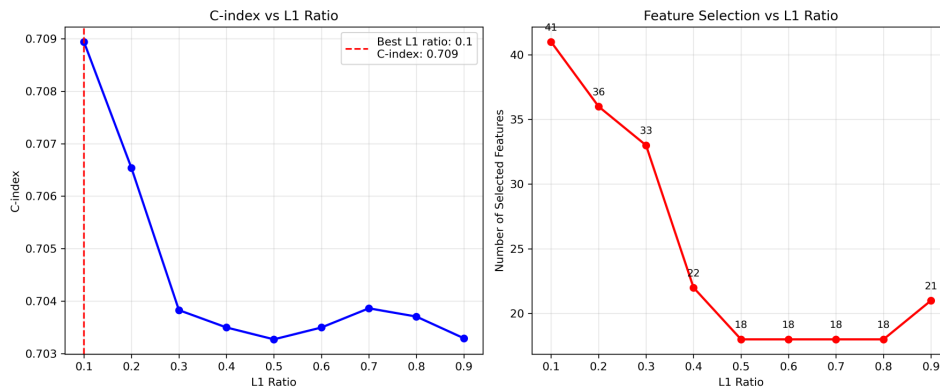


Figure 5-5 Grid Search Results of ElasticNet

ElasticNet selected 41 features (6.4% of the total). The results for ElasticNet variable importance remain largely consistent with those of Lasso-Cox and Ridge-Cox. ElasticNet, with its hybrid penalty, slightly improves prediction accuracy by incorporating other features while retaining the key predictors identified by Lasso. This is likely due to its ability to handle correlations between predictors more stably.

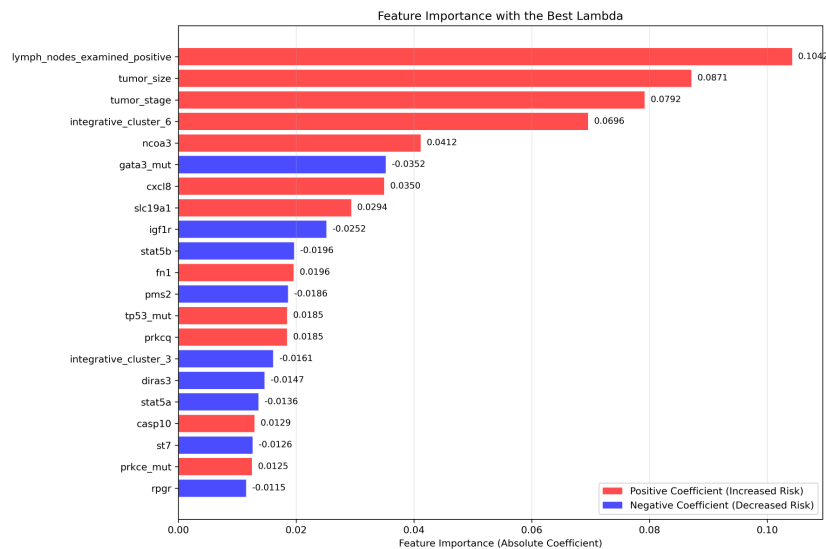


Figure 5-6 Feature Importance in ElasticNet

5.4 Comparative Analysis and Implications

Table 5-1 summarizes the comparison of the three regularized Cox models. Although the differences in the C-index are not significant, their performance hierarchy (ElasticNet > Ridge > Lasso) provides important methodological insights.

Generally, clinical genomic data typically has two characteristics: firstly, gene expression may be highly correlated; secondly, information is dispersed. Therefore, the L2 penalty in Ridge and ElasticNet effectively handles this multicollinearity and aggregates these weak, dispersed signals to obtain more robust predictions. While the Lasso model provides the most interpretable and concise model, it may be too aggressive in this context, discarding some features containing valuable incremental information. Furthermore, ElasticNet combines the advantages of Lasso and Ridge, extracting sufficient information and effectively achieving dimensionality reduction through feature selection, thus exhibiting higher stability and becoming the best-performing model among the three.

Table 5-1 Comparison of Regularized Cox Proportional Hazards Models

Model	Hyperparameters	Features Selected	Test C-index	Key Strengths
Lasso-Cox	$\lambda = 0.0560$	18 (2.8%)	0.703	High interpretability, strong feature selection.
Ridge-Cox	$\lambda = 658.3868$	637 (100%)	0.706	Handles multicollinearity, stable estimates.
ElasticNet-Cox	$\lambda = 0.4698, \alpha = 0.1$	41 (6.4%)	0.709	Balances feature selection with stability.

Biological and clinically, the consistency of the four key features, `lymph_node_status`, `tumor_size`, `integrative_cluster_6`, and `tumor_stage`, across all models strongly validates the central role of these clinical factors in breast cancer prognosis. Furthermore, the repeated selection of specific genes, such as `CXCL8` (involved in inflammation and angiogenesis), `GATA3_mut` (a key transcription factor in breast biology), and `tp53_mut` (genomic instability), highlights their potential as reliable genomic biomarkers, warranting further functional studies.

Additionally, ridge regression identified several important variables, such as `pam50_plus_claudin-low_subtype` (an important classification criterion for breast cancer), `neoplasm_histologic_grade` (whether cells are aggressive), and `chemotherapy`, revealing that these variables provide incremental information for survival prediction, offering direction for further clinical research.

5.5 Model Validation and Risk Stratification

To visually assess the clinical utility of the prognostic models, patients in the test set were stratified into low-, medium-, and high-risk groups based on the predicted risk scores from each regularized Cox model. Kaplan-Meier survival curves were plotted for these groups.

As illustrated in figure 5-7, all three regularization methods successfully generated models that effectively stratified patients into distinct prognostic groups. The survival curves for the low-, medium-, and high-risk groups demonstrated clear separation, confirming that the models capture meaningful variations in patient survival.

However, the KM curves revealed nuanced differences in the quality of stratification. The curves produced by the Ridge-Cox and ElasticNet-Cox models show more pronounced and consistent separation across the entire follow-up period, particularly between the medium- and high-risk groups. In contrast, the separation between these two groups in the Lasso-Cox model is less distinct, especially in the early to mid-follow-up period.

This observation aligns with the models' inherent characteristics and their performance on the test set. The L1 penalty in Lasso, while excellent for creating a sparse and interpretable model, might be overly aggressive, potentially discarding features that provide incremental but consistent prognostic information. This can lead to a slightly less refined risk stratification. Conversely, the L2 penalty in Ridge and the combined penalty in ElasticNet handle multicollinearity more effectively and retain a broader set of predictive signals, even if weak or correlated. This allows them to produce a more stable and finely graduated risk score, which translates into more clearly separated survival curves.

In conclusion, while all regularized models are valid for high-dimensional risk stratification, Ridge and ElasticNet offer a marginal but perceptible advantage in the clarity and consistency of the risk groups they define, underscoring the value of their approach to handling complex genomic data.

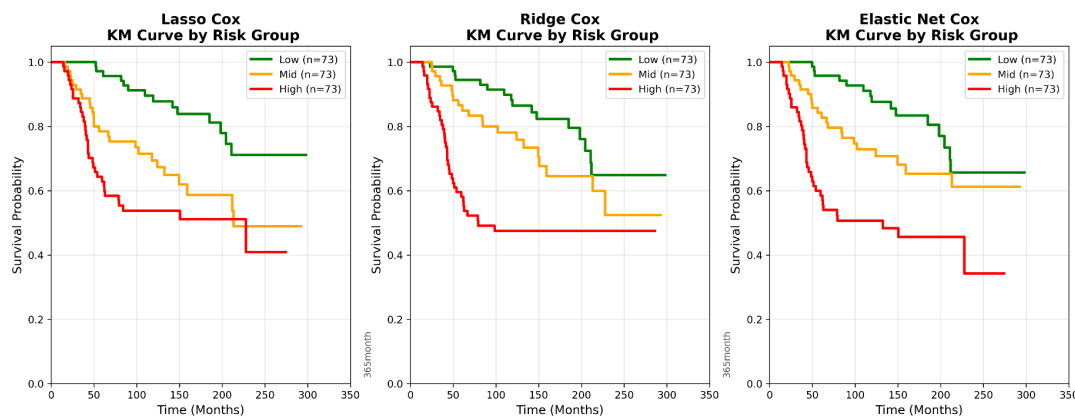


Figure 5-7 KM Curves for Three Models

6 Further Analysis: Ensemble Learning Models

To explore potential non-linear relationships and complex interactions among features that might not be fully captured by the linear Cox Proportional Hazards model, we employed several ensemble learning methods: Random survival forest, XGBoost and LightGBM.

6.1 Initial Model with Full Features Set

We first constructed a Random Survival Forest (RSF) model using all 637 features. This model was tuned via grid search, resulting in optimal parameters of `n_estimators=200` and `max_depth=5`.

RSF's feature importance ranking (figure 6-1) notably identified `lymph_nodes_examined_positive`, `tumor_size`, `tumor_stage`, `chemotherapy`, and key genes like `stat5b` among its top predictors. This high degree of overlap with the most influential variables from the regularized Cox models robustly cross-validates the biological and clinical relevance of these core prognostic factors.

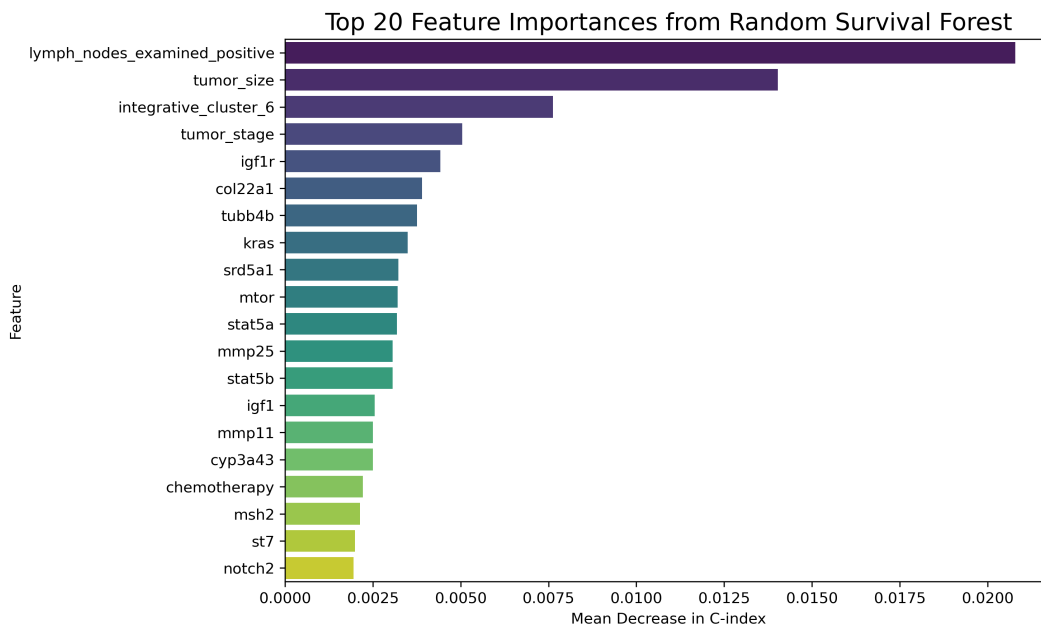


Figure 6-1 RSF Feature Importance

However, this initial model exhibited a critical flaw: severe overfitting. It achieved a near-perfect training C-index of 0.898 but generalized poorly to the test set, with a C-index of only 0.695. This significant performance gap confirmed that directly applying complex, high-capacity models to the high-dimensional feature space without robust feature selection leads to learning noise rather than generalizable signals.

6.2 Regularized Feature Selection Coupled with Ensemble Learning

To mitigate overfitting while harnessing the power of ensemble methods, we implemented a two-stage hybrid framework. This involved using the parsimonious yet informative feature subsets previously selected by the regularized Cox models—the 18 features from Lasso-Cox and the 41 features from ElasticNet-Cox—as the input for a suite of ensemble learners. We constructed and evaluated four hybrid models:

- Lasso-RF: Random Survival Forest trained on the 18 Lasso-selected features.
- ElasticNet-RF: Random Survival Forest trained on the 41 ElasticNet-selected features.
- ElasticNet-XGBoost: XGBoost model trained on the 41 ElasticNet-selected features.
- ElasticNet-LightGBM: LightGBM model trained on the 41 ElasticNet-selected features.

All models were rigorously tuned via grid search to ensure optimal performance. The performance of these hybrid models, evaluated on the independent test set, is summarized in Table 6-1.

Table 6-1 Performance of Hybrid Ensemble Learning Models

Model	Feature Set	Test C-index
RSF	637 features	0.695
Lasso-RSF	18 features	0.711
ElasticNet-RSF	41 features	0.713
ElasticNet-XGBoost	41 features	0.718
ElasticNet-LightGBM	41 features	0.721

This comparative analysis yielded several key findings. First of all, all hybrid models demonstrated robust generalization, with test C-indices significantly higher than the overfit RSF model and comparable to or better than the linear Cox models. This confirms that using regularized models as feature filters is a highly effective strategy for controlling overfitting in ensemble learning applied to high-dimensional data.

Secondly, a clear performance gradient emerged among the ensemble techniques when using the same feature set: LightGBM (0.721) > XGBoost (0.718) > Random Survival Forest (0.713). This suggests that the gradient boosting framework (XGBoost, LightGBM), with its sequential error-correction mechanism, is more effective than bagging (RSF) at capturing the complex predictive patterns in this curated feature space. LightGBM's top performance is likely due to its efficient histogram-based algorithm and advanced techniques like Gradient-based One-Side Sampling (GOSS).

Finally, comparing the two RSF-based hybrids, the model using the larger ElasticNet feature set (41 features) slightly outperformed the one using the sparser Lasso set (18 features).

This indicates that the additional features retained by ElasticNet's less aggressive penalty contained valuable, non-redundant prognostic information that the ensemble model could leverage.

6.3 Conclusion and Implication

Methodologically, it demonstrates that a hybrid pipeline, combining the feature selection of regularized linear models with the superior pattern recognition capability of modern gradient boosting algorithms (particularly LightGBM) represents a powerful and robust strategy for prognostic modeling in high-dimensional genomic data. This approach successfully balances interpretability with predictive accuracy.

Biologically and clinically, the consistent prominence of key clinical features (`lymph_nodes_examined_positive`, `tumor_size`, `tumor_stage`) across all modeling paradigms, from linear Cox to ensemble models, underscores their fundamental and non-negotiable role in determining breast cancer prognosis. The fact that these factors remain paramount even when models are granted the flexibility to discover complex non-linear relationships powerfully reinforces their utility as the cornerstone of clinical risk assessment.

7 Conclusions

Leveraging the high-dimensional METABRIC dataset, this study has presented multiple prognostic models for prediction in breast cancer. Among the models, the ElasticNet-Cox model offered an optimal balance of interpretability and performance (C-index=0.709), while the hybrid ElasticNet-LightGBM model achieved the highest predictive accuracy (C-index=0.721). The core finding demonstrates that while traditional clinical factors such as lymph node status, tumor size, and tumor stage remain the most potent prognostic indicators, a consensus set of genomic features such as *CXCL8* and *GATA3_mut* was consistently identified across all modeling paradigms, underscoring their fundamental role in breast cancer progression.

The implications of this work can be delineated across three domains. First, this research provides a validated blueprint for analyzing high-dimensional biomedical data. Genomic data possesses two key characteristics: firstly, the data is high-dimensional and the variables are highly correlated, necessitating regularization as a crucial prerequisite for complex modeling, which is essential for mitigating overfitting. Secondly, in genomics, predictive signals are often scattered, and regularization methods like ridge and L2-regularized ElasticNet, capable of aggregating these weak but correlated signals, offer advantages. Furthermore, the superior performance of the hybrid pipeline—combining the feature selection advantages of regularized linear models with the ability of gradient boosters to capture complex interaction effects, ultimately demonstrates powerful predictive capabilities, establishing a robust and effective strategy for future prognostic studies.

Second, this study systematically revealed multiple prognostic factors for breast cancer. Among them, the L1-regularized model helps us quickly identify the key dominant factors driving prognostic differences, providing a foundation for mechanistic research and the development of tools to simplify clinical decision-making. The L2-regularized model, on the other hand, is heuristic, helping to capture signals with weaker effects but potentially incremental predictive value. This contributes to revealing underdeveloped potential genes or biomarkers, providing clues for subsequent biological function validation and pathogenic mechanism exploration.

Finally, the developed models hold tangible potential for enhancing clinical decision-making. The robust risk stratification model achieved by Kaplan-Meier curves could be refined into a clinical tool for personalized prognosis. By accurately identifying high-risk patients, clinicians could recommend more aggressive or targeted treatment strategies upfront. Conversely, low-risk patients might be spared unnecessary overtreatment, improving their quality of life.

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